

2. Row 1 Leading entry = 1 in Column 1

$\therefore$  The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is vacuously True.

3. Row 2 Column 1 = 0.

$\therefore$  The condition that for each column having a leading entry, all positions below the leading entry position must be 0 is True.

4. The condition that each leading entry must be 1 is True.

5. The condition that for each column having a leading entry, all positions above the leading entry position must be 0 is vacuously True.

$\therefore$  The matrix  $\begin{bmatrix} 1 & -3 & 2 \\ 0 & 0 & 0 \end{bmatrix}$  is in rref.

By defn, there are no pivots in Columns 2 and 3.  $\therefore$  By defn,  $x_2$  and  $x_3$  are free variables.

The corresponding equation is:  $x_1 - 3x_2 + 2x_3 = 0$

In parametric vector form, the <sup>solution set</sup> equation is:  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$

$$x_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \square$$

$$11. \begin{bmatrix} 1 & -4 & -2 & 0 & 3 & -5 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Row 1 = Row 1 -  
3xRow 3

$$\begin{bmatrix} 1 & -4 & -2 & 0 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Row 1 = Row 1 + } 2 \times \text{Row 2} \rightarrow \begin{bmatrix} 1 & -4 & 0 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$